

Project Write-Up

IITB Insti-Assist

RAG-Powered AI Assistant for IIT Bombay

1. Chosen Scope and Rationale

The project focuses on building an AI assistant for IIT Bombay's academic information, including course registration, grading policies, the academic calendar, and examination rules.

Problem. General-purpose LLMs do not possess institution-specific knowledge and may hallucinate answers about IIT Bombay's academic regulations, such as grading policies or registration deadlines.

Rationale. Academic policies require high factual accuracy. Using a Retrieval-Augmented Generation (RAG) pipeline allows the assistant to answer only from official documents, improving reliability while minimizing hallucinations.

2. Data Sources Used

The knowledge base consists of five official (simulated/scraped) documents:

1. IITB UG Academic Rulebook
2. Grading Policy & CPI Guidelines
3. Autumn Semester Academic Calendar

These documents provide information on academic rules, grading, registration, examinations, and important deadlines.

3. Chunking Strategy and Justification

The documents were split using the `RecursiveCharacterTextSplitter` with:

Parameter	Value
Chunk Size	500 characters
Chunk Overlap	50 characters
Separators	["\n\n", "\n", ".", " "]

A chunk size of 500 characters keeps individual academic rules together, while the 50-character overlap preserves context across chunk boundaries. The embeddings were generated using `all-MiniLM-L6-v2`, providing efficient semantic retrieval with FAISS.

4. Known Limitations and Future Improvements

Current Limitations

- PDF tables often lose their structure during text extraction, reducing retrieval accuracy.
- The assistant is stateless and cannot remember previous user queries.

Future Improvements

- Use advanced PDF parsers (e.g., `Unstructured` or `Camelot`) for better table extraction.
- Add conversational memory using LangChain's `ConversationBufferWindowMemory`.
- Allow users to upload personal course syllabi through the Streamlit interface for dynamic RAG.